KAIZEN MATHS

Lesson starter

Newton's Second Law: F = ma

Systems and Inclined Planes

Name

S1

Two particles of masses 4 kg and 4 kg are connected by a light string on a smooth horizontal surface. A horizontal force of 8 N pulls the 4 kg particle. Find the acceleration and the tension in the string.

S2

A person of mass 60 kg stands in a lift accelerating upwards at 0.4 m s^{-2}. Find the normal reaction from the floor. Take g = 9.8 m s^{-2}.

S3

A particle of mass 10 kg slides down a smooth plane inclined at 25° to the horizontal. Find its acceleration. Take g = 9.8 m s^{-2}.

S4

A particle of mass 10 kg slides down a rough plane inclined at 35° to the horizontal. The coefficient of friction is 0.1. Find its acceleration. Take g = 9.8 m s^{-2}.

Show your working

KAIZEN MATHS

Lesson starter

Newton's Second Law: F = ma

Systems and Inclined Planes

Name

S1

Two particles of masses 4 kg and 4 kg are connected by a light string on a smooth horizontal surface. A horizontal force of 8 N pulls the 4 kg particle. Find the acceleration and the tension in the string.

S2

A person of mass 60 kg stands in a lift accelerating upwards at 0.4 m s^{-2}. Find the normal reaction from the floor. Take g = 9.8 m s^{-2}.

S3

A particle of mass 10 kg slides down a smooth plane inclined at 25° to the horizontal. Find its acceleration. Take g = 9.8 m s^{-2}.

S4

A particle of mass 10 kg slides down a rough plane inclined at 35° to the horizontal. The coefficient of friction is 0.1. Find its acceleration. Take g = 9.8 m s^{-2}.

MEC-BF46-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Newton's Second Law: F = ma

Systems and Inclined Planes

Name _____

EA1

Two particles of masses 2 kg and 3 kg are connected by a light string on a smooth horizontal surface. A horizontal force of 4 N pulls the 2 kg particle. Find the acceleration and the tension in the string.

EA2

A person of mass 80 kg stands in a lift accelerating upwards at 0.4 m s^{-2}. Find the normal reaction from the floor. Take g = 9.8 m s^{-2}.

Show your working

KAIZEN MATHS

Exit ticket

Newton's Second Law: F = ma

Systems and Inclined Planes

Name _____

EB1

A particle of mass 6 kg slides down a smooth plane inclined at 40° to the horizontal. Find its acceleration. Take g = 9.8 m s^{-2}.

Show your working

KAIZEN MATHS

Exit ticket

Newton's Second Law: F = ma

Systems and Inclined Planes

Name _____

EA1

Two particles of masses 2 kg and 3 kg are connected by a light string on a smooth horizontal surface. A horizontal force of 4 N pulls the 2 kg particle. Find the acceleration and the tension in the string.

EA2

A person of mass 80 kg stands in a lift accelerating upwards at 0.4 m s^{-2}. Find the normal reaction from the floor. Take g = 9.8 m s^{-2}.

MEC-BF46-L3 · EA

Show your working

KAIZEN MATHS

Exit ticket

Newton's Second Law: F = ma

Systems and Inclined Planes

Name _____

EB1

A particle of mass 6 kg slides down a smooth plane inclined at 40° to the horizontal. Find its acceleration. Take g = 9.8 m s^{-2}.

MEC-BF46-L3 · EB

Show your working

Teacher answers & working

Newton's Second Law: F = ma · Systems and Inclined Planes

MEC-BF46-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Two particles of masses 4 kg and 4 kg are connected by a light string on a smooth horizontal surface. A horizontal force of 8 N pulls the 4 kg particle. Find the acceleration and the tension in the string.

Answer:

$a = 1\ m\ s^{-2}$, $T = 4\ N$

Working:

Treat both particles as one system to find the acceleration.

Total mass = 4 + 4 = 8 kg

Resultant external force = total mass x acceleration

$8 = 8a$

Divide both sides by the total mass.

$a = \frac{8}{8} = 1\ m\ s^{-2}$

Use the second particle alone to find the tension.

$T = m_2 a$

$T = 4 \times 1$

$T = 4\ N$

$a = 1\ m\ s^{-2}$, $T = 4\ N$.

S2

A person of mass 60 kg stands in a lift accelerating upwards at 0.4 m s^{-2}. Find the normal reaction from the floor. Take g = 9.8 m s^{-2}.

Answer:

612 N

Working:

Take upward as positive.

$R - mg = ma$

$R - (60 \times 9.8) = 60 \times 0.4$

$R - 588 = 24$

$R = 588 + 24$

$R = 612\ N$

Normal reaction = 612 N.

Teacher answers & working

Newton's Second Law: F = ma · Systems and Inclined Planes

MEC-BF46-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

A particle of mass 10 kg slides down a smooth plane inclined at 25° to the horizontal. Find its acceleration. Take g = 9.8 m s^{-2}.

Answer:

$4.14\ m\ s^{-2}$

Working:

Resolve the weight parallel to the plane.

Component of weight down the plane = mg sin 25°

Component down the plane = 10 × 9.8 × sin 25°

The plane is smooth, so there is no friction.

Resultant force down the plane = ma

$mg \sin 25° = ma$

$g \sin 25° = a$

$a = 9.8 \times \sin 25° = 4.14\ m\ s^{-2}$

Acceleration = 4.14 m s^{-2}.

S4

A particle of mass 10 kg slides down a rough plane inclined at 35° to the horizontal. The coefficient of friction is 0.1. Find its acceleration. Take g = 9.8 m s^{-2}.

Answer:

$4.82\ m\ s^{-2}$

Working:

Resolve the weight parallel and perpendicular to the plane.

Component down the plane = mg sin 35° = 10 × 9.8 × sin 35° = 56.21 N

Normal reaction = mg cos 35° = 10 × 9.8 × cos 35° = 80.28 N

Friction = μR = 0.1 × 80.28 = 8.03 N

Take down the plane as positive.

Resultant force down the plane = component of weight - friction

Resultant force = 56.21 - 8.03 = 48.18 N

$F = ma$

$48.18 = 10a$

Divide both sides by the mass.

$a = \frac{48.18}{10} = 4.82\ m\ s^{-2}$

Acceleration = 4.82 m s^{-2}.

Teacher answers & working

Newton's Second Law: F = ma · Systems and Inclined Planes

MEC-BF46-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

Two particles of masses 2 kg and 3 kg are connected by a light string on a smooth horizontal surface. A horizontal force of 4 N pulls the 2 kg particle. Find the acceleration and the tension in the string.

Answer:

$a = 0.8\ m\ s^{-2}$, $T = 2.4\ N$

Working:

Treat both particles as one system to find the acceleration.

Total mass = 2 + 3 = 5 kg

Resultant external force = total mass x acceleration

$4 = 5a$

Divide both sides by the total mass.

$a = \frac{4}{5} = 0.8\ m\ s^{-2}$

Use the second particle alone to find the tension.

$T = m_2 a$

$T = 3 \times 0.8$

$T = 2.4\ N$

$a = 0.8\ m\ s^{-2}$, $T = 2.4\ N$.

EA2

A person of mass 80 kg stands in a lift accelerating upwards at 0.4 m s^{-2}. Find the normal reaction from the floor. Take g = 9.8 m s^{-2}.

Answer:

816 N

Working:

Take upward as positive.

$R - mg = ma$

$R - (80 \times 9.8) = 80 \times 0.4$

$R - 784 = 32$

$R = 784 + 32$

$R = 816\ N$

Normal reaction = 816 N.

Teacher answers & working

Newton's Second Law: F = ma · Systems and Inclined Planes

MEC-BF46-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

A particle of mass 6 kg slides down a smooth plane inclined at 40° to the horizontal. Find its acceleration. Take g = 9.8 m s^{-2}.

Answer:

6.3 m s^{-2}

Working:

Resolve the weight parallel to the plane.

Component of weight down the plane = mg sin 40°

Component down the plane = 6 × 9.8 × sin 40°

The plane is smooth, so there is no friction.

Resultant force down the plane = ma

$mg \sin 40° = ma$

$g \sin 40° = a$

$a = 9.8 \times \sin 40° = 6.3$ m s^{-2}

Acceleration = 6.3 m s^{-2}.